

BCSE306

Artificial Intelligence

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Objectives

- ❑ To impart artificial intelligence principles, techniques and its history.
- ❑ To assess the applicability, strengths, and weaknesses of the basic knowledge representation, problem solving, and learning methods in solving engineering problems
- ❑ To develop intelligent systems by assembling solutions to concrete computational problems

Course Outcome

1. Evaluate Artificial Intelligence (AI) methods and describe their foundations.
2. Apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation and learning.
3. Demonstrate knowledge of reasoning and knowledge representation for solving real world problems.
4. Analyze and illustrate how search algorithms and planning play vital role in problem solving.

Module:1	Introduction	6 hours
Introduction- Evolution of AI, State of Art -Different Types of Artificial Intelligence- Applications of AI-Subfields of AI-Intelligent Agents- Structure of Intelligent Agents- Environments		
Module:2	Problem Solving based on Searching	6 hours
Introduction to Problem Solving by searching Methods-State Space search, Uninformed Search Methods – Uniform Cost Search, Breadth First Search- Depth First Search-Depth-limited search, Iterative deepening depth-first, Informed Search Methods- Best First Search, A* Search		
Module 3	Local Search and Adversarial Search	5 hours
Local Search algorithms – Hill-climbing search, Simulated annealing, Genetic Algorithm, Adversarial Search: Game Trees and Minimax Evaluation, Elementary two-players games: tic-tac-toe, Minimax with Alpha-Beta Pruning.		
Module:4	Logic and Reasoning	8 hours
Introduction to Logic and Reasoning -Propositional Logic-First Order Logic-Inference in First Order Logic- Unification, Forward Chaining, Backward Chaining, Resolution.		
Module:5	Uncertain Knowledge and Reasoning	5 hours
Quantifying Uncertainty- Bayes Rule -Bayesian Belief Network- Approximate Inference in Bayesian networks		
Module:6	Planning	7 hours
Classical planning, Planning as State-space search, Forward search, backward search, Planning graphs, Hierarchical Planning, Planning and acting in Nondeterministic domains – Sensor-less Planning, Multiagent planning		
Module:7	Communicating, Perceiving and Acting	6 hours
Communication-Fundamentals of Language -Probabilistic Language Processing -Information Retrieval- Information Extraction-Perception-Image Formation- Object Recognition.		

Text Book
1. Russell, S. and Norvig, P. 2015. Artificial Intelligence - A Modern Approach, 3 rd Edition, Prentice Hall.

Reference Books
1. K. R. Chowdhary, Fundamentals of Artificial Intelligence, Springer, 2020.
2. Alpaydin, E. 2010. Introduction to Machine Learning. 2 nd Edition, MIT Press.

ONLINE COURSE

- <https://nptel.ac.in/courses/106106226>
- https://onlinecourses.nptel.ac.in/noc23_cs09/preview

FIELDS OF ARTIFICIAL INTELLIGENCE(AI)

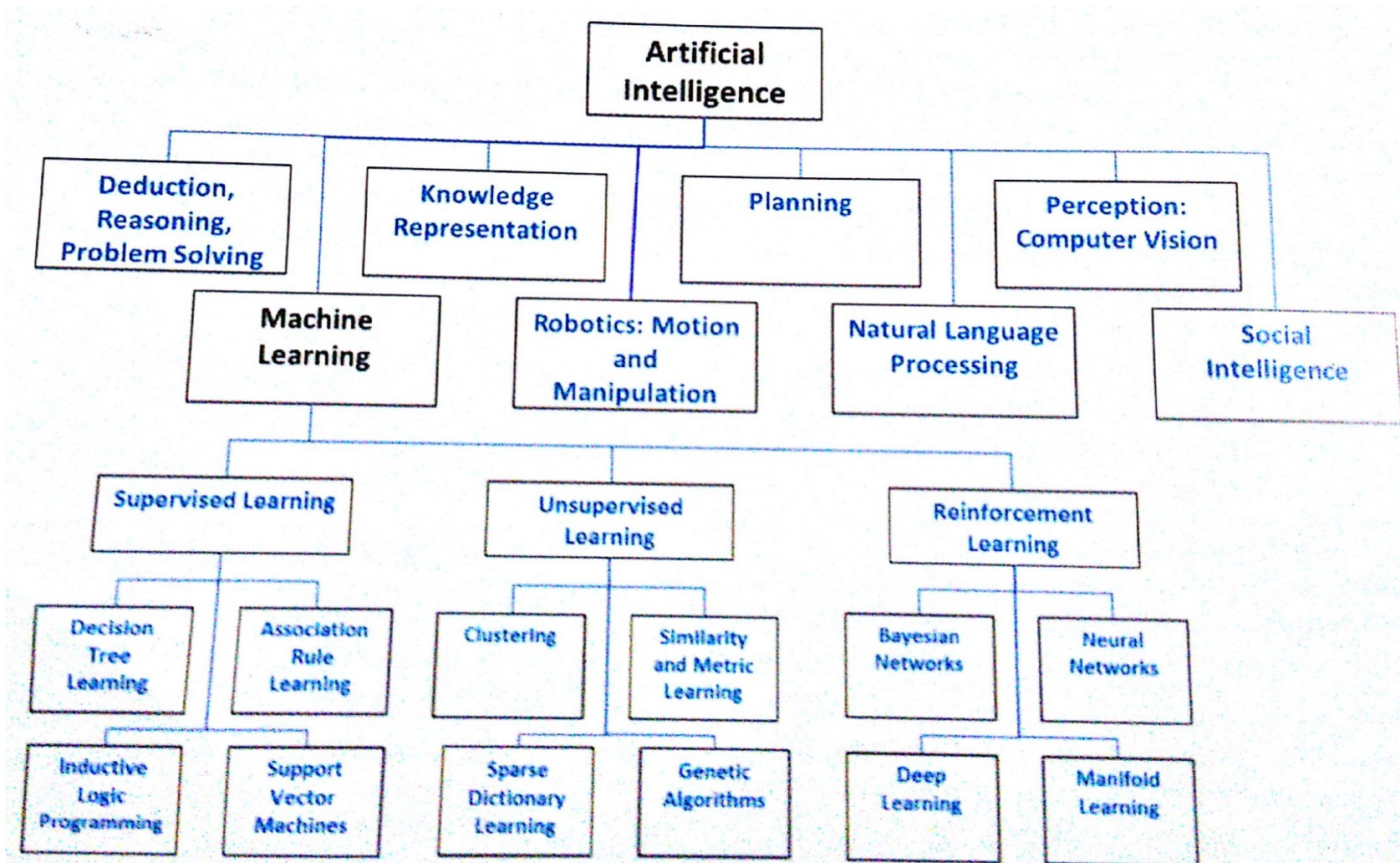


Fig. 2: Fields of Artificial Intelligence^[4]

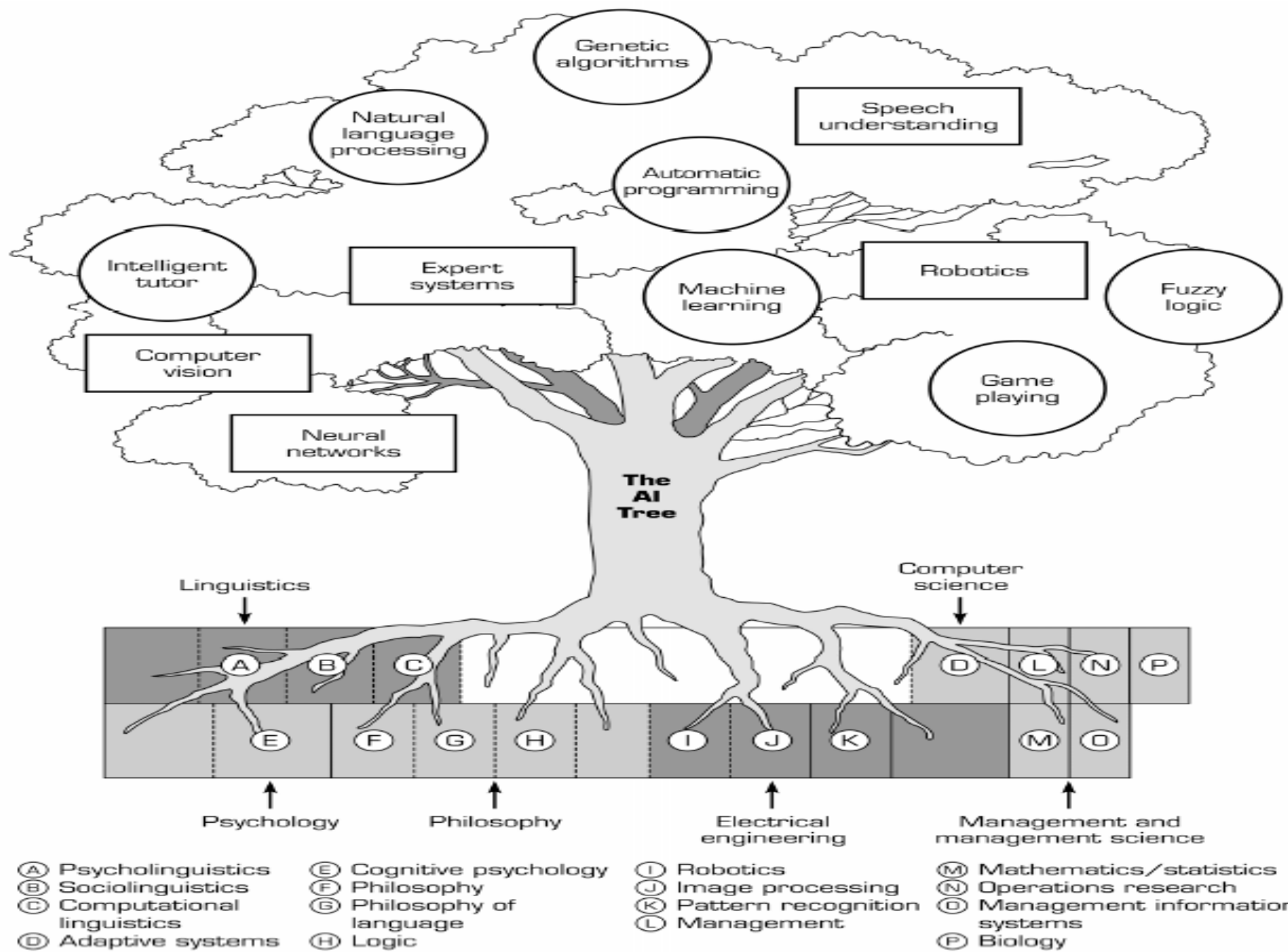
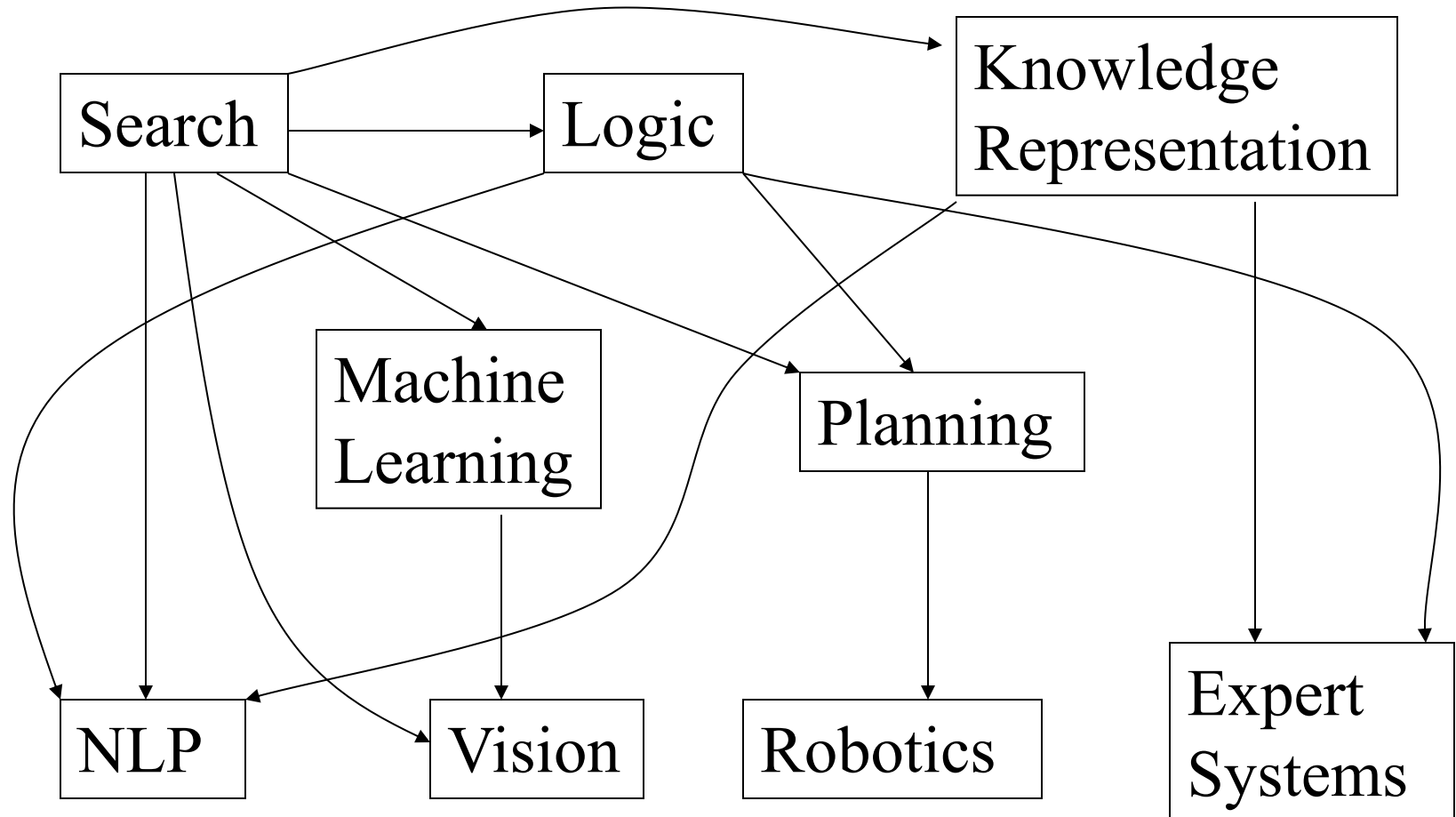


FIGURE 12.2 The Disciplines and Applications of AI

Areas of AI and Some Dependencies



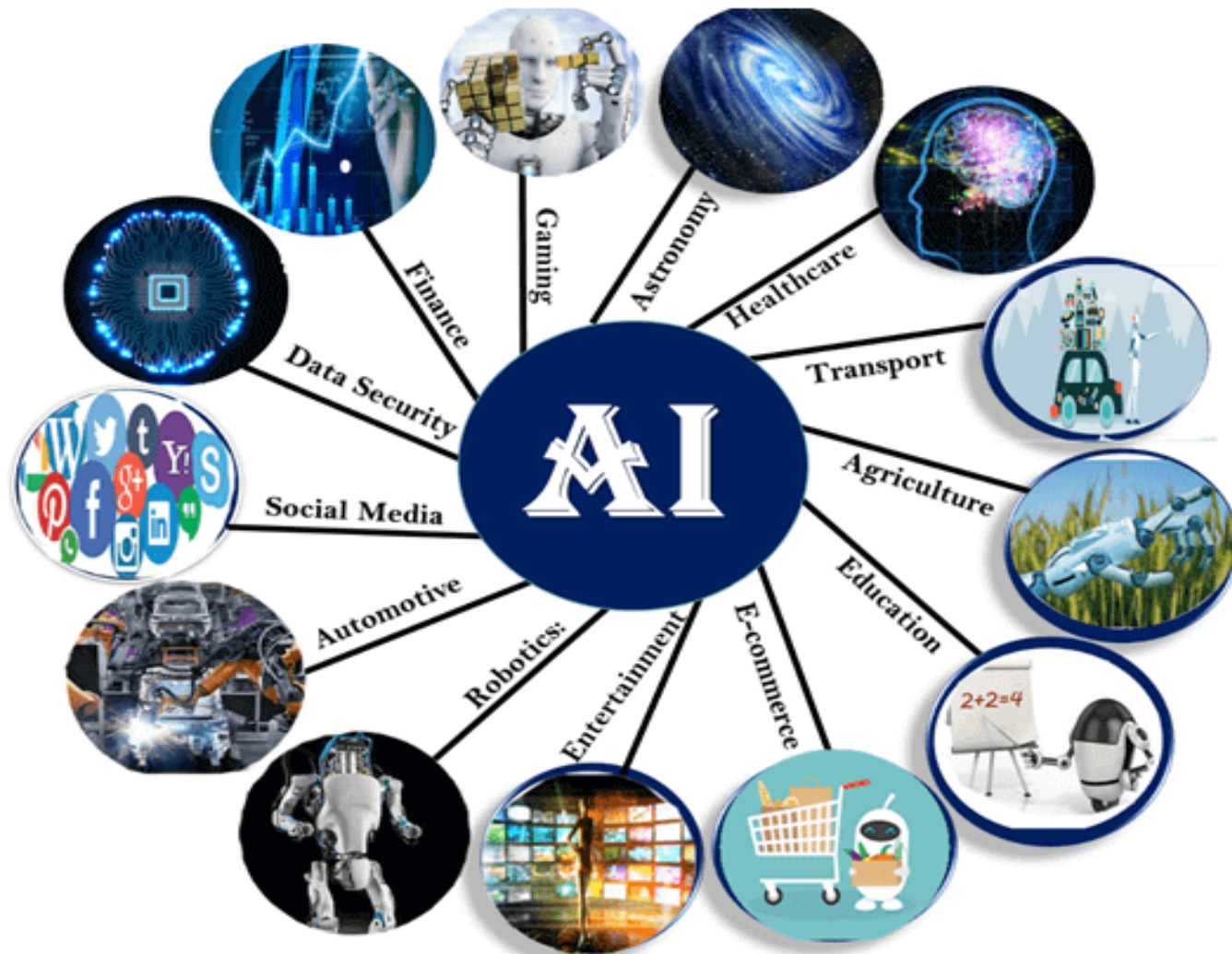
What Can AI do?

- AI automates repetitive learning and discovery through data.
- AI adds intelligence
- AI adapts through progressive learning algorithms
- AI analyzes more and deeper data
- AI achieves incredible accuracy
- AI gets the most out of data

Refer -> <https://www.youtube.com/watch?v=G-zyTIZQYpE>

Applications of Artificial Intelligence

Applications



AI : Applications

Agriculture	Mathematics
Architecture	Military
Art	Music
Astronomy	Scientific discovery
Bioinformatics	Story writing
Email classification	Telecommunications
Engineering	Telephone services
Finance	Transportation
Fraud detection	Tutoring systems
Information retrieval	Video games
Law	Web search engines

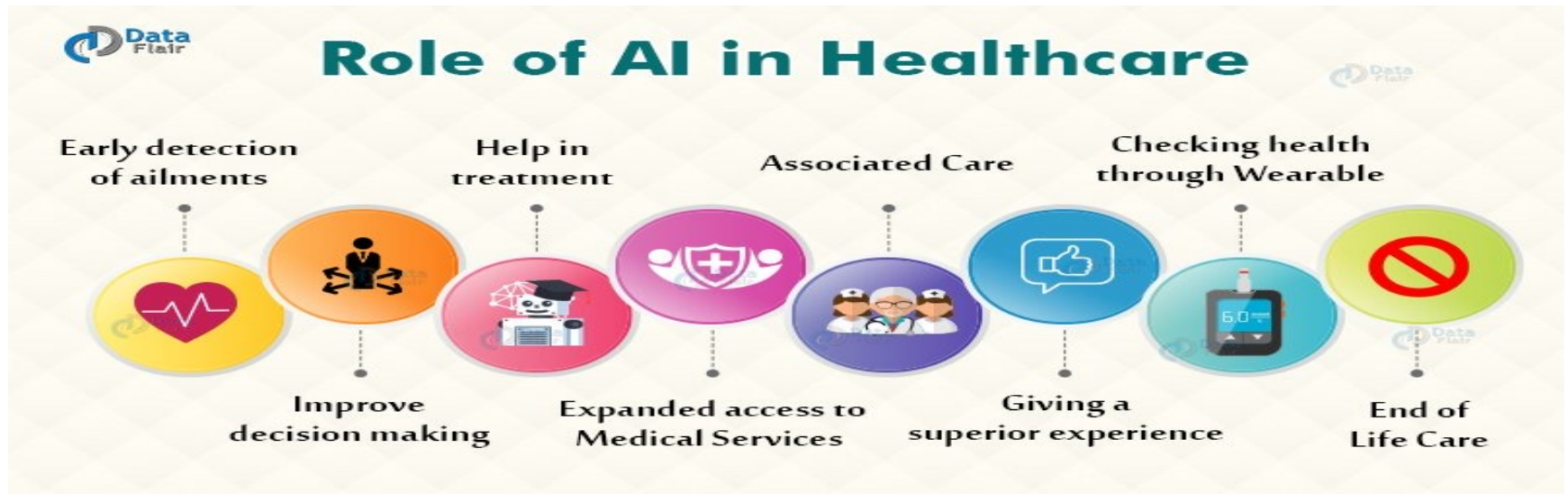
AI applications in Astronomy

- Artificial Intelligence can be very **useful in solving complex universe problems.**
- AI technology can help us **understand the universe such as how it works, its origin,** etc.
- To take on the challenges, astronomers are turning to machine learning and artificial intelligence algorithms to build new tools to rapidly search for the next big breakthroughs.



AI Applications in Healthcare

- In the last, five to ten years, AI has become **more advantageous for the healthcare industry** and going to **have a significant impact on this industry**.
- Healthcare Industries are applying **AI to make a better and faster diagnosis than humans**.
- AI can **help doctors with diagnoses** and can **inform when patients are worsening** so that medical help can reach the patient before hospitalization.



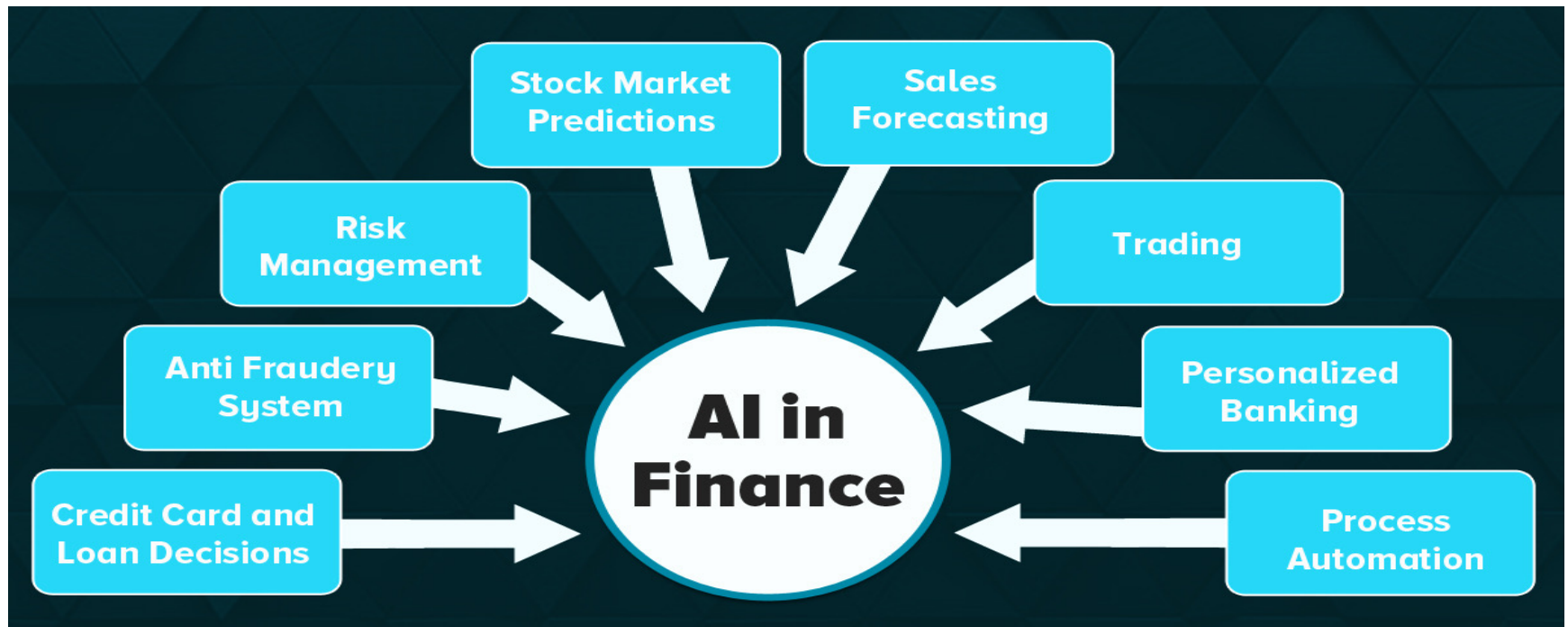
AI Applications in Gaming

AI can be used for gaming purposes **which can play strategic games like chess**, where the machine needs to think of a large number of possible places.



AI Applications in Finance

- AI and finance industries are the **best matches for each** other.
- The finance industry is implementing **automation, chatbots, adaptive intelligence, algorithm trading, and machine learning** into financial processes.



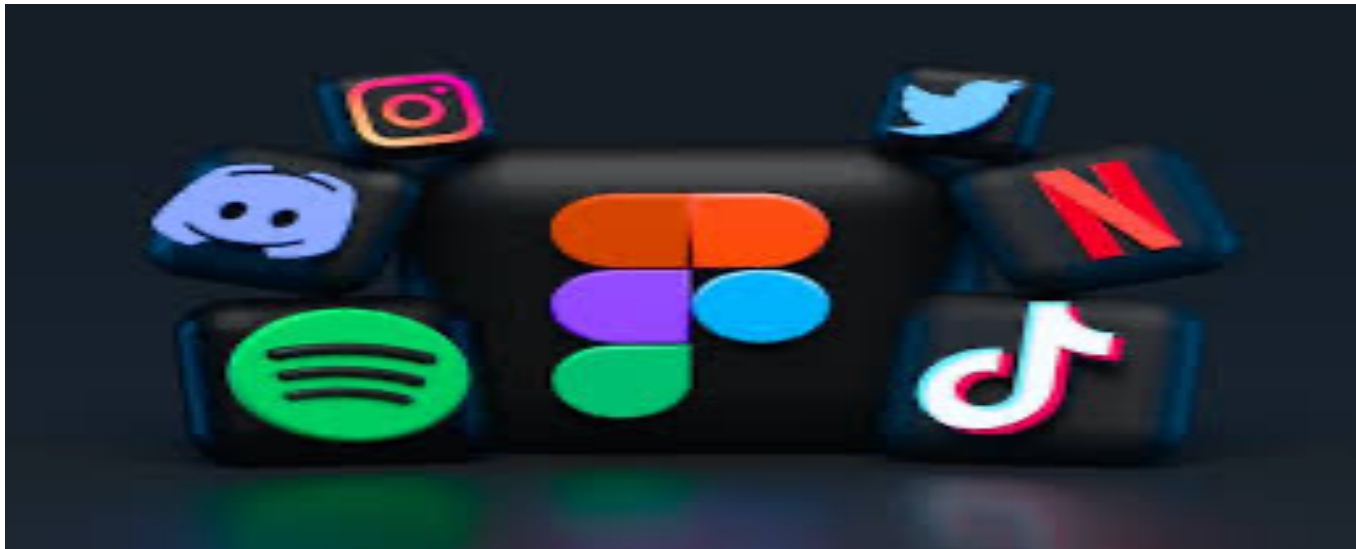
AI Applications in Data Security

- Data security is crucial for every company, and cyber-attacks are growing in the digital world.
- AI can be used to make your data more safe and secure.
- Some examples such as the AEG bot, and AI2 Platform are used to determine software bugs and cyber-attacks in a better way.



AI Applications in Social Media

- **Social Media** sites such as Facebook, Twitter, and Snapchat contain billions of user profiles, which need to be stored and managed in a very efficient way.
- AI can **organize and manage massive amounts of data**.
- AI can analyze lots of data to **identify the latest trends, hashtags, and requirements** of different users.



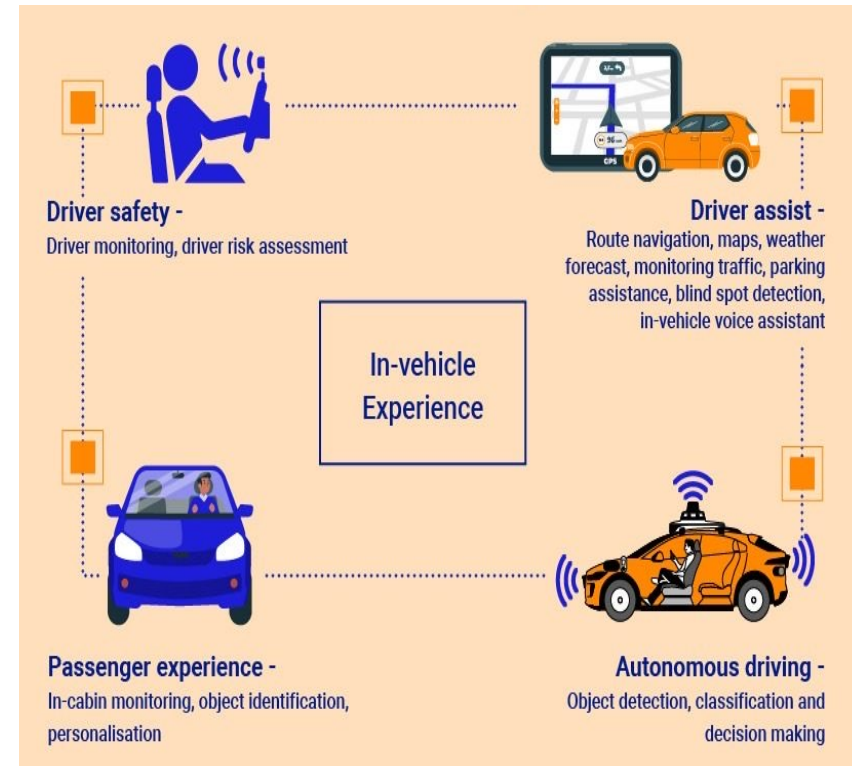
AI Applications in Travel & Transport

- AI is becoming **highly demanding** for travel industries.
- AI is capable of doing various travel - related work such as **making travel arrangements to suggesting the hotels, flights, and best routes to the customers.**
- Travel industries are using **AI-powered chatbots** which can make **human-like interaction with customers** for better and fast response.



AI Applications in the Automotive Industry

- Some Automotive industries are using AI to **provide virtual assistant to their user** for better performance.
- Such as **Tesla** has introduced **TeslaBot**, an intelligent virtual assistant.
- Various Industries are currently working to develop **self-driven cars** that can make your journey more safe and secure.



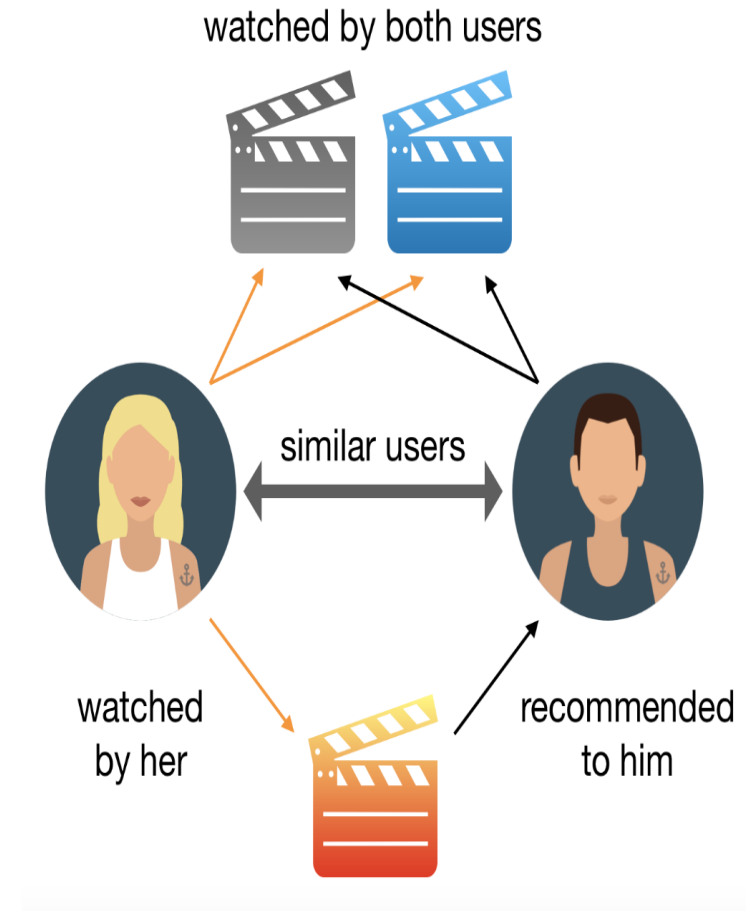
AI Applications in Robotics

- Artificial Intelligence has a **remarkable role in Robotics**.
- Usually, **general robots are programmed** such that they can **perform some repetitive task**, but with the help of AI, we can create intelligent robots which can perform tasks with their own experiences without pre-programmed.
- Humanoid Robots are best examples for AI in robotics, recently the **intelligent Humanoid robot named as Erica and Sophia** has been developed which can talk and behave like humans.



AI Applications in Entertainment

- We are currently using some **AI-based applications** in our daily lives with some entertainment services such as **Netflix or Amazon**.
- With the help of ML/AI algorithms, these services **show the recommendations** for programs or shows.



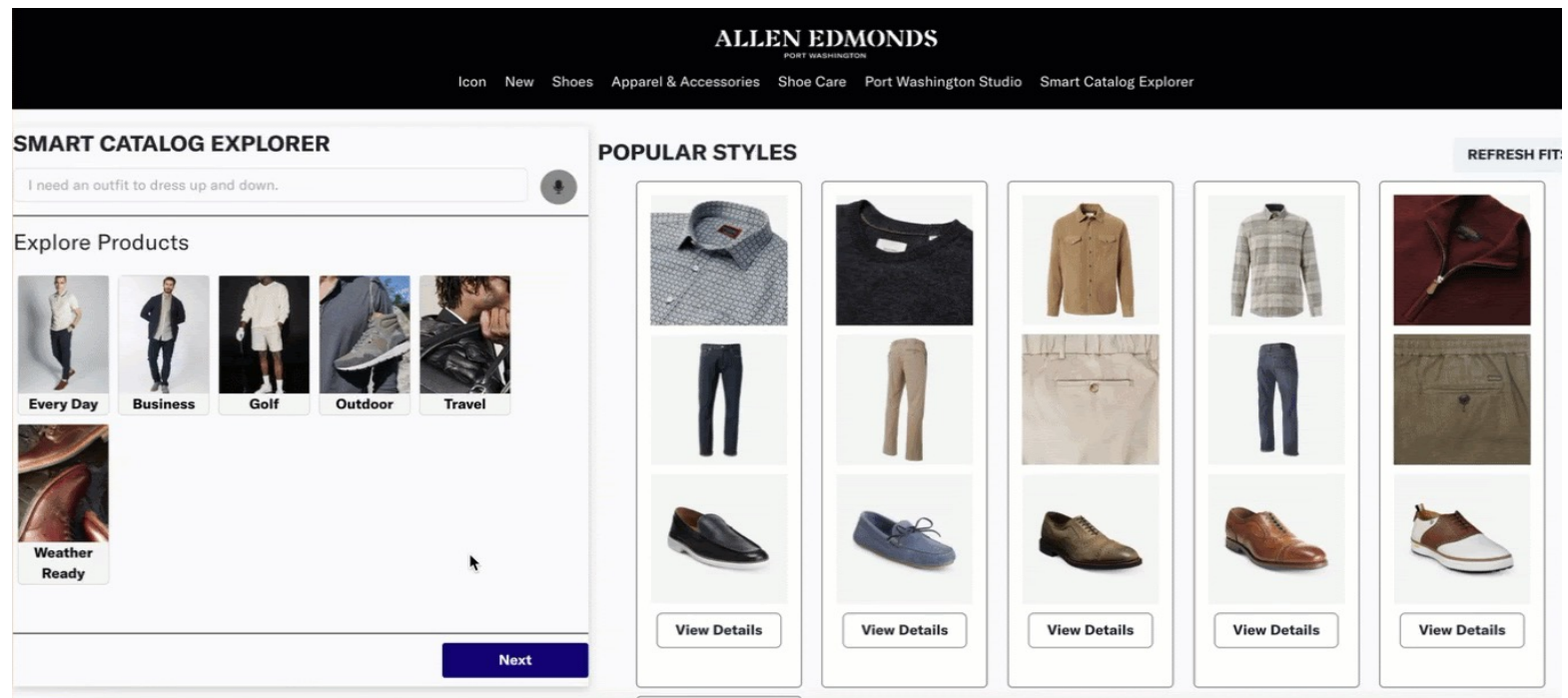
AI Applications in Agriculture

- Agriculture is an area that **requires various resources, labor, money, and time for the best result.**
- Nowadays agriculture is becoming digital, and **AI is emerging in this field.**
- Agriculture is applying AI in **agriculture robotics, solid and crop monitoring, and predictive analysis.**
- AI in agriculture can be **very helpful for farmers.**



AI Applications in E-commerce

- AI is providing a **competitive edge** to the e-commerce **industry**, and it is becoming **more demanding** in the e-commerce **business**.
- AI is helping shoppers **discover associated products** with **recommended sizes, colors, or even brands**.



AI Applications in education

- AI can automate grading so that the tutor can have more time to teach.
- AI Chatbots can communicate with students as a teaching assistant.
- AI in the future can work as a personal virtual tutor for students, which will be accessible easily at any time and any place.



Other application areas

- **Bioinformatics**
 - Gene expression data analysis
 - Prediction of protein structure
- **Text classification, and document sorting**
 - Web pages, e-mails
 - Articles in the news
- **Video, image classification**
- **Music composition, picture drawing**
- **Natural Language Processing**

